claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-a3917e155f343ce93.jsonl`
- SHA-256: `d0370460b8c8913b9be80b14435f3403da8beefdec869e071170382c5e5ed749`
- Source modified: `2026-06-06T17:39:49+00:00`
- Imported at: `2026-07-05T16:48:28+00:00`
- project: `wf_6237b110-d58`
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`

Transcript

1. user (2026-06-06T17:37:51.098Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):
- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:
1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ?

especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

Claim under review

"Hit-frame detection (per-shot temporal localization, useful as a segmentation primitive) is driven primarily by player pose/court keypoints rather than only the shuttle trajectory ? shuttlecock flying direction is predicted from player keypoint sequences with over 92% accuracy."

Source: https://arxiv.org/pdf/2307.16000 (primary)
Supporting quote: "The proposed approach in this study comprises several automated procedures like rally-wise video trimming, player and court keypoints detection, shuttlecock flying direction prediction, and hit-frame detection."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

refuted=true if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.
refuted=false ONLY if: claim is well-supported, current, and source quality matches claim strength.
Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-06T17:37:57.045Z)

I'll verify this claim by examining the source paper and searching for corroborating/contradicting evidence.

3. assistant (2026-06-06T17:38:00.091Z)

Now let me fetch the primary source paper to examine the claim.

4. user (2026-06-06T17:38:08.381Z)

I apologize, but I cannot extract meaningful information from the provided content. The text appears to be a corrupted or improperly encoded PDF file containing primarily binary data and compression artifacts rather than readable academic content.

The stream contains FlateDecode-compressed data that, when decoded, does not produce legible text that would allow me to identify:
- The paper's title, year, and venue
- The pipeline components you mentioned
- Accuracy metrics for shuttlecock direction prediction
- Hit-frame detection methodology
- Specific technical details about inputs and performance numbers

To provide the detailed information you're requesting (including exact accuracy figures and method descriptions), I would need access to a properly formatted and readable version of the paper?either as a standard PDF that renders correctly or as plain text.

If you have an alternative version of this document or can provide the paper in a different format, I'd be happy to analyze it and extract the specific technical details you need.

[Binary content (application/pdf, 1.5MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\tool-results\webfetch-1780766977975-dnjzkl.pdf]

5. user (2026-06-06T17:38:08.874Z)

Web search results for query: "shuttlecock flying direction prediction player keypoints badminton hit-frame detection accuracy arxiv 2307.16000"

Links: [{"title":"[2307.16000] Automated Hit-frame Detection for Badminton Match Analysis","url":"https://arxiv.org/abs/2307.16000"},{"title":"Automated Hit-frame Detection for Badminton Match Analysis - ADS","url":"https://ui.adsabs.harvard.edu/abs/2023arXiv230716000C/abstract"},{"title":"Transformer on Shuttlecock Flying Direction Prediction for Hit-frame Detection","url":"https://www.researchgate.net/publication/372785149_Transformer_on_Shuttlecock_Flying_Direction_Prediction_for_Hit-frame_Detection"},{"title":"Automated Hit-frame Detection for Badminton Match Analysis","url":"https://arxiv.org/pdf/2307.16000"},{"title":"Future Prediction of Shuttlecock Trajectory in Badminton Using Player?s Information","url":"https://www.mdpi.com/2313-433X/9/5/99"},{"title":"Future Prediction of Shuttlecock Trajectory in Badminton Using Player?s Information","url":"https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10219238/"},{"title":"(PDF) Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera","url":"https://www.researchgate.net/publication/382039345_Enhancing_Badminton_Game_Analysis_An_Approach_to_Shot_Refinement_via_a_Fusion_of_Shuttlecock_Tracking_and_Hit_Detection_from_Monocular_Camera"}]

Based on the search results for arxiv paper 2307.16000, here's what I found:

This research aims to advance sports analysis in badminton, systematically detecting hit-frames automatically from match videos using modern deep learning techniques.

The proposed approach comprises several automated procedures like rally-wise video trimming, player and court keypoints detection, shuttlecock flying direction prediction, and hit-frame detection.

Regarding accuracy metrics, the study achieved 99% accuracy on shot angle recognition for video trimming, over 92% accuracy for applying player keypoints sequences on shuttlecock flying direction prediction, and reported evaluation results for rally-wise video trimming and hit-frame detection.

The paper specifically addresses using player keypoint sequences to predict shuttlecock flying direction, which is a key component of the automated hit-frame detection system for badminton match analysis.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-06T17:38:12.555Z)

The PDF was binary-unreadable via WebFetch, but the search gave me strong signal. Let me get the abstract page and a detailed read of the methodology to verify how hit-frame detection actually works versus the flying-direction prediction.

7. user (2026-06-06T17:38:16.853Z)

The server returned HTTP 403 Forbidden.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

8. user (2026-06-06T17:38:19.781Z)

Paper Summary

Full Abstract

"Sports professionals constantly under pressure to perform at the highest level can benefit from sports analysis, which allows coaches and players to reduce manual efforts and systematically evaluate their performance using automated tools. This research aims to advance sports analysis in badminton, systematically detecting hit-frames automatically from match videos using modern deep learning techniques. The data included in hit-frames can subsequently be utilized to synthesize players' strokes and on-court movement, as well as for other downstream applications such as analyzing training tasks and competition strategy. The proposed approach in this study comprises several automated procedures like rally-wise video trimming, player and court keypoints detection, shuttlecock flying direction prediction, and hit-frame detection. In the study, we achieved 99% accuracy on shot angle recognition for video trimming, over 92% accuracy for applying player keypoints sequences on shuttlecock flying direction prediction, and reported the evaluation results of rally-wise video trimming and hit-frame detection."

Reported Accuracy Numbers

- **Shot angle recognition:** 99%
- **Shuttlecock flying direction prediction:** Over 92%
- **Hit-frame detection:** Evaluation results are mentioned but specific accuracy not provided in abstract

Shuttlecock Direction Prediction

Yes, the abstract explicitly states: "over 92% accuracy for applying player keypoints sequences on shuttlecock flying direction prediction"

Hit-frame Detection Approach

The abstract does not specify whether hit-frame detection relies on player pose/court keypoints or shuttle trajectory. It only mentions that the system includes "player and court keypoints detection" and "hit-frame detection" as separate components.

9. assistant (2026-06-06T17:38:23.284Z)

I now have the verbatim abstract. Let me try to read the locally-saved PDF to verify the hit-frame detection mechanism specifically, since that's the load-bearing part of the claim.

10. user (2026-06-06T17:38:25.582Z)

```
Exit code 2
/usr/bin/bash: eval: line 1: unexpected EOF while looking for matching `"'
```

11. user (2026-06-06T17:38:30.973Z)

```
Traceback (most recent call last):
  File "<string>", line 1, in <module>
    import pdfplumber; pdf=pdfplumber.open('/mnt/c/Users/avidu/.claude/projects/C--Users-avidu-Projects-badminton-highlight-indexer/42211f80-2169-4025-ad9d-99c1aa29ff70/tool-results/webfetch-1780766977975-dnjzkl.pdf'); print(chr(10).join((p.extract_text() or '') for p in pdf.pages))
                           ~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\site-packages\pdfplumber\pdf.py", line 98, in open
    stream = open(path_or_fp, "rb")
FileNotFoundError: [Errno 2] No such file or directory: '/mnt/c/Users/avidu/.claude/projects/C--Users-avidu-Projects-badminton-highlight-indexer/42211f80-2169-4025-ad9d-99c1aa29ff70/tool-results/webfetch-1780766977975-dnjzkl.pdf'
```

12. assistant (2026-06-06T17:38:33.884Z)

```
pdfplumber is on Windows Python, not WSL Python. Let me use the Windows path directly.
```

13. user (2026-06-06T17:38:34.888Z)

```
pdftoppm failed: Command 'pdftoppm' not found or is in an unsafe location (current directory)
```

14. user (2026-06-06T17:38:42.624Z)

```
35771
```

15. assistant (2026-06-06T17:38:50.880Z)

```
Extracted 35K of text. Let me search for the hit-frame detection methodology specifically.
```

16. user (2026-06-06T17:38:51.556Z)

```
1:Automated Hit-frame Detection for Badminton Match Analysis
12:identifyhit-framesinrawmatchvideos.Byoutliningame-
14:thodicalstrategyforautomaticallyidentifyinghit-framese-
16:systematicallydetectinghit-framesautomaticallyfrommatch
quencesfrompubliclyavailablematchvideos,thisworkin-
18:cluded in hit-frames can subsequently be utilized to syn- can facilitate downstream
applications dependent on infor-
19:thesize players? strokes and on-court movement, as well as mation included in hit-frames,
such as stroke classification
25:tlecock flying direction prediction, and hit-frame detection.
26:Inthestudy,weachieved99%accuracyonshotanglerecog- Focusingontrajectory-
relatedstudies,(LiuandWang2022)
27:nition for video trimming, over 92% accuracy for applying reconstructs the 3D trajectory of
shots from unlabeled
28:player keypoints sequences on shuttlecock flying direction videos using various neural
networks and a per-shot tra-
30:videotrimmingandhit-framedetection.
31:model-based trajectory estimation method based on linear
33:Introduction NetV2,toforecastthetrajectoryofthetinyandfast-moving
62:fication in compressed videos using shuttlecock trajectory.
63:matchprogressedandoffertacticalinformation;hit-frames,
69:tion.Additionally,hit-framespossessessentialdata,includ-
82:encoder-decoderextractorsandaposition-awarefusionnet- Algorithm1:Hit-
framegeneratingalgorithm.
84:As for studies related to footwork and movements, Output:Hit-framesetHit.
105:nents such as trajectory construction, stroke classification,
109:the information within the hit-frames (discussed in ), like
113:knowledge,nopreviousworkpredictstheshuttlecock?sfly- 16: hit?HitFrameDetection(s)
118:then detect hit-frames according to the sequence. Such an 20:
kps?DetectPlayerKeypoints(frame)
120:cationsrequiringhit-framesinformation.
130:andpinpointhit-frames.Aalgorithm1depictstheprocessof O
(likeline14),itindicatestheendofarally,thealgo-
131:all the procedures. The hit-frame generating algorithm in-
rithmpredictstheflyingdirectionsoftheshuttlecockus-
133:a set of hit-frame indices denoted by Hit. Lines 1 to 3 de- it to s. Then, it proceeds to
detect hit-frames hit using
134:finetheshotangleconstantsO (representingother)andH
HitFrameDetection(illustratedin)andaddsittotheHit
142:(forstoringdetectedhit-frameindices),bothinitiallyempty.
249:are scaled to a range of 0 and 1 (Deng et al. 2009). Other tect human and court keypoint
sequences. These keypoint
279:amodelthatextendstheFasterR-CNN(Girshick2015)by Identifying the shuttlecock?s flying
direction is essential
292:processcanbedefinedas: model that leverages keypoint sequences to predict shuttle-
296:For the shuttlecock flying direction prediction task, we
297:K =KeypointRCNN(I ),K ?RM×17×2 built a keypoint sequence dataset (KSeq dataset) with 12
308:ure3(a)),wetrainedaCourtR-CNNforcourtkeypointsde- keypoint sequences. For the ground truth,
```

we invited sev-
311:exceptitpredictedonlysixkeypoints,includingupperright, flying direction in the corresponding video frame. Further-
331:where xt stands for the rally-wise keypoint sequences, Xt
389:of two sub-layers. The first is a multi-head self-attention Algorithm2:Hit-Framedetectingalgorithm.
391:forward network. Around the two sub-layers, residual con- Output:Hit-framessethit.
405:Hit-FrameDetection
407:Hit-frame,whicharedefinedastheframethataplayerhits 9: elseifprevious direction=Bthen
411:tlecock. Such hit-frames can be identified by detecting to- 13: endif
412:ken shifts in the shuttlecock flying direction sequence. Al- 14: elseifprevious direction=U then
413:gorithm2depictshowwedevisedhit-framedetection. 15: ifdirection=Bthen
419:the previous flying direction encountered during the itera- 21: previous direction?direction
421:where the identified hit-frame indices will be stored. The 23: returnhit.
423:tlecock flying direction in the input sequence s. If the cur-
429:hit-frames.
431:isB orU,ittellsthataplayerstartstherally;therefore, 3. Canhit-framesbepreciselyidentifiedthroughflyingdi-
432:thealgorithmidentifiesthisframeasahit-frame.Itcalls rectionchanges?
435:2. IfthepreviousdirectionwasB andthecurrentdirection wisevideotrimmingmodule,transformer,andhit-framede-
437:court strikes the shuttlecock, causing a flying direction
445:tected hit-frame indices, representing potential badminton fyingshotangles.
452:minton by systematically detecting hit-frames from match
459:Table3:Numbersofcorrectlytrimmed,extratrimmed,miss Table 6: Confusion matrix of the hit-frame detecting mod-
472:andendframenumber,whichweadoptedintheevaluation actualhit-framenumbers,andP istheprediction.Theresult
473:stage.Table3showsthenumberofralliestrimmedandthe ofhit-framedetectionisshowninTable6.FromTable6,we
474:actual rally count, while Table 4 shows the confusion ma- caninferthatthehit-framedetectingmodulecandistinguish
475:trix of the SA-CNN. From Tables 3 and 4, we acknowl- mosthit-frameswithintherange±15.
477:rally-related videos. Such alimitation is due to the varying Hit-FrameApplications
479:Numerouscurrentstudiesonbadmintoninvolvehit-frames.

[Showing results with pagination = limit: 60]

17. assistant (2026-06-06T17:38:55.837Z)

This is the crux. The key lines are 405-437, describing the Hit-Frame Detection mechanism. Let me read them in full context.

18. user (2026-06-06T17:38:56.603Z)

279 amodelthatextendstheFasterR-CNN(Girshick2015)by Identifying the shuttlecock?s flying direction is essential
280 addingabranchforpredictingsegmentationmasksoneach for downstream analysis since it implies information about
281 Region of Interest in parallel with the existing branch for when the players hit the shuttlecock. In a badminton rally,
282 classification and bounding box regression. The Keypoint shots will occur several times until it ends, when a fault
283 R-CNNmodelisbasedontheMaskR-CNNmethodandex- happens, or when the shuttlecock hits the floor. Therefore,
284 pectsitsinputtobeanimageofallsizeswhiletheoutputis we conclude the flying status of the shuttlecock into the
285 alistofdictionaries.Eachdictionaryprovidesthepredicted following three state classes: steady, flying toward the bot-
286 box,label,confidencescore,andaninstance?skeypoints.In tom court, and flying toward the upper court, then formu-
287 our case, we only utilized the instance?s keypoints, which latetheshuttlecockflyingdirectionpredictionasasequence
288 include the following 17 human parts: nose, left eye, right labeling task with tokens S,

B, and U representing the
eye, left ear, right ear, left shoulder, right shoulder, left el- three classes respectively. We assume the player keypoint
bow, right elbow, left wrist, right wrist, left hip, right hip, sequence is essential in predicting the shuttlecock?s flying
left knee, right knee, left ankle, right ankle. The detection direction. Consequently, we proposed a novel transformer
processcanbedefinedas: model that leverages keypoint sequences to predict shuttle-
cockflyingdirectionsequences.
I ?R1080×1920
rally
For the shuttlecock flying direction prediction task, we
K =KeypointRCNN(I),K ?RM×17×2 built a keypoint sequence dataset (KSeq dataset) with 12
rally
uniquemen?ssinglesmatchesfromtheBWFchannelasthe
whereI denotestherally-relatedframe,K denotesthe
rally datasource.Tenmatcheswereusedtotrainthetransformer
detectedhumankeypointcoordinates,andM representsthe
model, while two were for testing. The sequence data in
countofhumansdetected.
the KSeq dataset comprises multiple frame-wise keypoint
Court Keypoints Detection As Keypoint R-CNN in- pairs (2 players). Table 1 shows the information of KSeq
evitablyreturnsallthehumansitdetected(asshowninFig- dataset.Intotal,thedatasetiscomposedof1134rally-wise
ure3(a)),wetrainedaCourtR-CNNforcourtkeypointsde- keypoint sequences. For the ground truth, we invited sev-
tection to filter players? information. The output format of eralbadmintonexpertstolabeleachkeypointpairwithone
theCourtR-CNNissimilartothatoftheKeypointR-CNN, ofthementionedthreeclassesaccordingtotheshuttlecock
exceptitpredictedonlysixkeypoints,includingupperright, flying direction in the corresponding video frame. Further-
upper left, middle right, middle left, bottom right, and bot- more,paddingtoken(tokenPad)isappliedtoensureeach
tom left points (as shown in Figure 3(b)). Since we focus keypointsequencehasthesamelengthduringtraining,and
only on men?s singles matches, the detection target of the thekeypointswillbenormalizedwiththedataset?saverage
modelwillbethesinglescourt. andstandarddeviationbeforeenteringthetransformer.
K =CourtRCNN(I),K ?R1×6×2 Wetrainedthetransformerfor100epochswithanAdam
court rally court optimizer, whose learning rate is set as 1e-5 and decayed
whereK standsforthecourt?skeypointscoordinates. 90% after the seventieth epoch. The training process is de-
court
(a) (b) (c)
Figure3:(a)IllustratestheKeypointR-CNNmodel?soutputwithoutfilteringtheinstances.(b)showstheoutputoftheCourt
R-CNNmodel.(c)IllustratestheoutcomeoftheintegratedmethodofKeypointR-CNNandCourtR-CNN.
finedasfollows: Inputs
xt =(K ···K)
player(1) player(F)
Xt =(xt ···xt),xt ?RF×2×17×2
1 N
Player-Wise Multi-layer Multi-layer
y?t =Transformer(Xt),y?t ?RN×F×C
Projection Perceptron Perceptron
where xt stands for the rally-wise keypoint sequences, Xt
representsthetrainingdatabatch,ytistheshuttlecockdirec-
tionsequence,andF representstheframesequencelength.
The loss function Lt for transformer training is defined as
follows: Keypoints Feed Forward
Keypoints
Multi-Head
yt ={yt···yt },yt ?{S,B,U,Pad} (4) Encoder Attention Add & Norm

1 N n
Lt =lt(y?t,yt)={l 1 t···l n t}T (5) Add & Norm Dropout = 0.1
(cid:26)
1, ifx?=ignoreindex
Ignore(x)= (6)
0, else
? exp (cid:16) y?t (cid:17) ? Keypoints Multi-layer
l
n
t =?log?
(cid:80)C exp
n
(cid:16)
,
y
y
?
n t
t
(cid:17)?·Ignore(y
n
t) (7) Decoder Perceptron
c=1 n,c
Softmax
lt(y?t,yt)= (cid:88) N l n t (8)
(cid:80)N Ignore(yt)
n=1 n=1 n Outputs
InthelossfunctionLt,N spansthebatchdimensionaswell
as F. During training, N = 1,F = 600,C = 4, and the
Figure4:Thearchitectureoftheadoptedtransformermodel.
paddingtokenPadissetastheignoreindex,soitisignored
anddoesnotcontributetotheinputgradientwhencalculat-
ingtheloss.Figure4depictsthetransformer?sstructure. layerisdefinedasfollows:
Player-WiseProjection Aplayer-wiseprojectionlayeris FC(x)=ReLU(W FC ·x+b FC)
built to initially project the input keypoints to a higher di- H(i) =FC (cid:0) FC (cid:0) xt[:,i,:,:] (cid:1)(cid:1) ,i?{1,2}
mension before feeding the transformer sequences of nu- player
(cid:16) (cid:17)
merouspairsofkeypoints.Theplayer-wiseprojectionlayer xt =Concatenate H(1) ,H(2)
projected player player
comprisestwomulti-layerperceptronblockswithtwofully
connected layers. Before entering the layer, the keypoint whereW ,b isthelearnableweightmatricesandbias
FC FC
data of the two players will be momentarily separated, and ofthefullyconnectedlayer,H(i) denotestheoutputofthe
following the projection procedure, they will be concate- player
i player-wise projection layer before concatenation, and
th
nated.Inourexperiment,thetransformermodelwithplayer- xt representstheprojectedkeypointsequencetensor.
wise projection layers outperforms the model without it, projected
supporting our hypothesis that this mechanism helps sepa- keypointEncoderandDecoder Theencoderstackcom-
rate the information of the two players rather than consid- prises several identical layers, each built based on the ar-
ering them as a single identity. The player-wise projection chitecture described in (Vaswani et al. 2017) and consists
of two sub-layers. The first is a multi-head self-attention Algorithm2:Hit-Framedetectingalgorithm.
layer; the second is a position-wise fully connected feed- Input:Shuttlecockflyingdirectionsequences.
forward network. Around the two sub-layers, residual con- Output:Hit-framessethit.
nection is proposed, followed by layer normalization and
1: LetconstsS ?0,B ?1,U ?2

394 dropout.Themodel?soutputsfromallsub-layershaveiden-
395 2: Letprevious direction?S
396 ticaldimensionstofacilitatetheseresidualconnections.The
397 3: Lethit??
398 transformermodelwebuiltfeatureseightencoderlayersand
399 4: foralldirection?sdo
400 eight heads for the attention mechanism; the dimension of
401 5: ifprevious direction?=directionthen
402 thefeed-forwardlayeris2048.
403 6: ifprevious direction=S then
404 7: h?FrameIndex(direction)
405 Hit-FrameDetection
406 8: hit.add(h)
407 Hit-frame,whicharedefinedastheframethataplayerhits 9: elseifprevious direction=Bthen
408 theshuttlecockandthuschangesitsflyingdirection,canbe 10: ifdirection=U then
409 utilized to retrieve data for downstream analysis since they 11: h?FrameIndex(direction)
410 possessinformationabouttheplayerswhentheyhittheshut- 12: hit.add(h)
411 tlecock. Such hit-frames can be identified by detecting to- 13: endif
412 ken shifts in the shuttlecock flying direction sequence. Al- 14: elseifprevious
direction=U then
413 gorithm2depictshowwedevisedhit-framedetection. 15: ifdirection=Bthen
414 Lines 1 to 3, the algorithm starts by defining three con- 16: h?FrameIndex(direction)
415 stants:S,B,andU,whichrepresentthestatesoftheshut- 17: hit.add(h)
416 tlecockflyingdirectionasfollows:steady,flyingtowardthe 18: endif
417 bottom court, and flying toward the upper court. The vari- 19: endif
418 ableprevious directionisinitializedtoS tokeeptrackof 20: endif
419 the previous flying direction encountered during the itera- 21: previous
direction?direction
420 tion. The set hit is also created and initialized as empty, 22: endfor
421 where the identified hit-frame indices will be stored. The 23: returnhit.
422 algorithm then enters a loop that iterates over each shut-
423 tlecock flying direction in the input sequence s. If the cur-
424 Table2:ConfusionmatrixoftheSA-CNN.
425 rentdirectiondiffersfromthepreviousdirectionwithinthe
426 loop,theshuttlecock?sflyingdirectionhaschanged.Theal- Accuracy Precision Recall F1-Score
427 gorithmthenchecksforspecificcasesoftransitionstodetect
428 0.9976 0.9903 1.0 0.9951
429 hit-frames.
430 1. IfthepreviousdirectionwasS andthecurrentdirection
431 isB orU,ittellsthataplayerstartstherally;therefore, 3. Canhit-
framesbepreciselyidentifiedthroughflyingdi-
432 thealgorithmidentifiesthisframeasahit-frame.Itcalls rectionchanges?
433 FrameIndex to get the frame index h corresponding to
Wewillanswerthesequestionsinthefollowingparagraphs,
434 thecurrentdirectionandaddsittothesethit. and the assessment and experiment of the SA-
CNN, rally-
435 2. IfthepreviousdirectionwasB andthecurrentdirection
wisevideotrimmingmodule,transformer,andhit-framede-
436 is U, or vice versa, it indicates the player at the bottom
tectingmodulearethemaintopicsofthissection.
437 court strikes the shuttlecock, causing a flying direction
438 transition (or vice versa). The algorithm calls FrameIn- SA-CNN

19. assistant (2026-06-06T17:39:05.169Z)

I now have the complete methodology. The claim is precisely supported by the paper. Let me verify the chain of logic carefully against the exact text.

The paper's pipeline, verbatim:
- **Shuttlecock flying direction prediction**: "We assume the player keypoint sequence is essential in predicting the shuttlecock's flying direction. Consequently, we proposed a novel transformer model that leverages keypoint sequences to predict shuttlecock flying direction sequences." Inputs are explicitly `xt = (K_player(1) ··· K_player(F))` ? human (player) keypoints from Keypoint R-CNN, filtered by Court R-CNN court keypoints. Reported accuracy: "over 92% accuracy for applying player keypoints sequences on shuttlecock flying direction

prediction." **No shuttlecock-trajectory tracker (e.g., TrackNet) is used as input** ? the direction is inferred from player pose, not from tracking the shuttle.
- **Hit-frame detection** (Algorithm 2): hit-frames are identified by "detecting token shifts in the shuttlecock flying direction sequence." That flying-direction sequence is itself produced from player keypoints. So hit-frame detection is downstream of, and driven by, the player-keypoint-based direction prediction ? not by a shuttle-trajectory tracker.

Let me do one independent cross-check on whether the "over 92%" figure and the keypoint-driven approach is corroborated, and confirm the paper date.

20. user (2026-06-06T17:39:16.614Z)

Web search results for query: ""Automated Hit-frame Detection for Badminton" Chen 2023 keypoint sequence transformer flying direction without TrackNet shuttle trajectory"

Links: [{"title":"GitHub - arthur900530/Automated-Hit-frame-Detection-for-Badminton-Match-Analysis","url":"https://github.com/arthur900530/Automated-Hit-frame-Detection-for-Badminton-Match-Analysis"},{"title":"Automated Hit-frame Detection for Badminton Match Analysis","url":"https://arxiv.org/pdf/2307.16000"},{"title":"Automated Hit-frame Detection for Badminton Match Analysis | alphaXiv","url":"https://www.alphaxiv.org/abs/2307.16000"},{"title":"[2307.16000] Automated Hit-frame Detection for Badminton Match Analysis","url":"https://arxiv.org/abs/2307.16000"},{"title":"Automated Hit-frame Detection for Badminton Match Analysis - ADS","url":"https://ui.adsabs.harvard.edu/abs/2023arXiv230716000C/abstract"},{"title":"Automated Hit-frame Detection for Badminton Match Analysis | DeepAI","url":"https://deepai.org/publication/automated-hit-frame-detection-for-badminton-match-analysis"},{"title":"Badminton Video Analysis based on Spatiotemporal and Stroke Features","url":"https://www.researchgate.net/publication/317159692_Badminton_Video_Analysis_based_on_Spatiotemporal_and_Stroke_Features"},{"title":"Statistical and AI-Based Techniques for Badminton Data Processing: Concepts, Applications, and Challenges | International Journal of Racket Sports Science","url":"https://revistaseug.ugr.es/index.php/IJRSS/article/view/36742"},{"title":"An All Deep System for Badminton Game Analysis","url":"https://arxiv.org/pdf/2308.12645"}]

Great! I found the research you're looking for. Here's what I found about the paper:

Automated Hit-frame Detection for Badminton Match Analysis (Chen/Chien et al., 2023)

This research aims to advance sports analysis in badminton, systematically detecting hit-frames automatically from match videos using modern deep learning techniques.

Key Technical Approach:

The work proposes a novel transformer that predicts shuttlecock direction sequences based on player keypoint sequence. The proposed approach comprises several automated procedures like rally-wise video trimming, player and court keypoints detection, shuttlecock flying direction prediction, and hit-frame detection.

Notable Results:

The study achieved 99% accuracy on shot angle recognition for video trimming, over 92% accuracy for applying player keypoints sequences on shuttlecock flying direction prediction.

Applications:

The data included in hit-frames can subsequently be utilized to synthesize players' strokes and on-court movement, as well as for other downstream applications such as analyzing training tasks and competition strategy.

You can find the full paper on arXiv at: https://arxiv.org/abs/2307.16000, and there's also a GitHub repository with the implementation: https://github.com/arthur900530/Automated-Hit-frame-Detection-for-Badminton-Match-Analysis

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

21. assistant (2026-06-06T17:39:32.027Z)

I have fully verified the claim against the primary source. Let me assess it adversarially against each checklist item.

Claim under review: "Hit-frame detection (per-shot temporal localization, useful as a segmentation primitive) is driven primarily by player pose/court keypoints rather than only the shuttle trajectory ? shuttlecock flying direction is predicted from player keypoint sequences with over 92% accuracy."

Adversarial assessment:

1. **Supported by quote?** The supporting quote merely lists pipeline stages. But I went to the full paper text and confirmed the deeper claim directly. Methodology section (lines 286-293): "We assume the player keypoint sequence is essential in predicting the shuttlecock's flying direction. Consequently, we proposed a novel transformer model that leverages keypoint sequences to predict shuttlecock flying direction sequences." Inputs (line 324-326): `xt = (K_player(1) ··· K_player(F))`, i.e., player keypoints from Keypoint R-CNN filtered by Court R-CNN ? no shuttle-tracking input. Hit-frame detection (Algorithm 2, lines 405-413): "Such hit-frames can be identified by detecting token shifts in the shuttlecock flying direction sequence" ? and that sequence is the keypoint-derived output. So hit-frame detection IS driven by player-pose-derived direction, not by a shuttle tracker. Abstract confirms "over 92% accuracy for applying player keypoints sequences on shuttlecock flying direction prediction."

2. **Contradicting evidence?** No credible contradiction found. GitHub repo and alphaXiv corroborate the keypoint-driven transformer approach. The paper itself states (line 113) "no previous work predicts the shuttlecock's flying direction" from keypoints ? a novelty claim, consistent.

3. **Source quality?** Primary source (arXiv paper, peer-reviewed-style with code on GitHub). Sufficient for the claim strength.

4. **Outdated?** 2023, current for this field. Fine.

5. **Marketing/cherry-pick?** No ? it's a research paper with confusion matrices and reported limitations.

One nuance worth flagging (potential overreach): The claim says "driven primarily by player pose/court keypoints **rather than only the shuttle trajectory**." This is accurate FOR THIS PAPER ? it deliberately avoids shuttle tracking and uses pose. But the claim's framing implies pose is generically superior/primary for hit-detection across the field, which the paper does NOT establish (it's one design choice; other papers like the "All Deep System" and the Shot Refinement fusion paper DO use shuttle tracking for hit detection). However, the claim as literally worded is scoped to "is driven primarily by..." with the 92% figure attributed correctly ? and that is exactly what the paper does. The claim is well-supported by the cited primary source.

The claim is accurate and well-supported. Not refuted.

22. user (2026-06-06T17:39:42.568Z)

Structured output provided successfully

23. assistant (2026-06-06T17:39:49.206Z)

Verification complete. The claim is NOT refuted ? it is well-supported by the primary source.

Summary: I confirmed against arXiv 2307.16000 (full PDF text extracted locally to `C:\Users\avidu\paper_text.txt` plus the verbatim abstract from the arXiv page) that hit-frame detection in this paper is indeed driven by player-pose/court keypoints, not by a shuttle tracker. The flying-direction predictor is a transformer over player keypoint sequences (inputs `xt = (K_player(1)...K_player(F))` from Keypoint R-CNN, filtered by Court R-CNN), reporting "over 92% accuracy," and hit-frames are derived by detecting token shifts in that pose-driven direction sequence (Algorithm 2). The 92% figure, the keypoint-driven mechanism, and the source quality/recency all check out. The only nuance (not enough to refute) is that the pose-only design is this paper's specific choice ? other badminton works do use shuttle tracking for hit detection ? but the claim as worded is faithful to the cited primary source.